

Kavish Shanilka

Electrical Engineering Undergraduate · University of Moratuwa, Sri Lanka

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About Me

Electrical engineering undergraduate at the University of Moratuwa passionate about research, intelligent systems, embedded hardware, and power electronics. Thrives in collaborative environments solving complex engineering challenges through innovative instrumentation and control systems.

Areas of Interest: Power Electronics, Embedded Systems, Computer Vision, Microgrids, Autonomous Systems, Digital Signal Processing, IoT, Robotics, Renewable Energy

Education

University of Moratuwa

B.Sc. Hons in Electrical Engineering (Undergraduate) – Cumulative GPA: 3.85 / 4.00

Feb. 2024 – Present

Moratuwa, Sri Lanka

Bandaranayake Central College

G.C.E. Advanced Level Examination (Physical Science Stream) – Z-Score: 2.6192

2011 – 2019

Veyangoda, Sri Lanka

- Achieved 3 As (Combined Mathematics, Physics, Chemistry) · **Island Rank: 131st** (District Rank: 12th)

Awards & Scholarships

Semi-Finalist – Spark Challenge

2025 – 2026

- Selected for semi-finals for designing **GridGuard**, an edge-coordinated smart energy management system.

Top 12 Finalist – GENESIZ '26 | KDU Robotics & Innovation Club

August 2026

- Achieved Top 12 Finalist with Team Stormforge for presenting **GridGuard** at the national innovation competition.

University Category Competitor – SLIOT Challenge 2026 | CSE Dept. (UoM), SLT-Mobitel, IESL

April 2026

- Competed as part of Team Neura in the national IoT innovation challenge with an automated hardware prototype.

Rotaractor Award | Rotaract Club of University of Moratuwa (Rotaract Mora)

2025

- Honored with the **Rotaractor Award (2025)** for dedicated leadership across community outreach and youth empowerment projects.

Projects

• GridGuard – Autonomous Edge-Coordinated Energy Management System 🔗

- Designed distributed IoT smart grid manager with Raspberry Pi central hub and ESP32 edge nodes for Spark Challenge & GENESIZ '26.
- Implemented non-invasive main DB power auditing and peak load-shifting algorithms, reducing electricity costs by > 30%.

• Smart Automatic Power Factor Corrector (APFC) Panel 🔗

- Engineered automated Power Factor Corrector panel featuring CT/VT sensing circuits and dynamic capacitor bank switching.
- Built custom LabVIEW dashboards with NI DAQ card and ESP32 OLED Telegram alerts for real-time status and safety monitoring.

• Smart Irrigation System Controller (MATLAB / Simulink) 🔗

- Modeled closed-loop plant dynamics and hysteresis controller logic for automated multi-zone water storage and routing.
- Integrated multi-zone moisture routing algorithms and fault detection logic to optimize water and energy consumption.

• Advanced Light Intensity Indicator (ALII) – Discrete Digital Hardware 🔗

- Designed discrete hardware signal processor without microcontrollers to monitor and average light intensity in real-time.
- Implemented analog filtering for 50–100 Hz harmonics and built flip-flop logic for time-windowed averaging (300–900 s).

Leadership & Experience

Senior Member – Public Relations Pillar | Administrative Committee

2026 – 2027

Institution of Engineers Sri Lanka (IESL) Student Chapter – University of Moratuwa

Moratuwa, Sri Lanka

- Appointed to 2026/2027 Admin Committee. Leading digital outreach, PR strategy, and industry communication channels connecting undergraduates with professional practice.

Editorial & Marketing Team Member

2025 – Present

Electrical Engineering Society (EESOC), University of Moratuwa

Moratuwa, Sri Lanka

- Contributing to departmental editorial publications, social media campaign artwork, and promotional strategies for flagship EESOC events.

Co-Director – Club Service

Rotaract Alumni University of Moratuwa

2025 – Present

Moratuwa, Sri Lanka

- Spearheading club service operations, fostering alumni-undergraduate engagement, fellowship events, and professional networking.

Active Member & Rotaractor Award Recipient

Rotaract Club of University of Moratuwa (Rotaract Mora)

2025 – Present

Moratuwa, Sri Lanka

- Honored with the **Rotaractor Award (2025)** for dedicated leadership across community outreach and youth empowerment projects.

Design Team Lead – Prakampana '26 | Project Chair – Project HOPE & CAST 4

2025 – 2026

Dept. of Electrical Engineering / Leo Club / Rotaract Club

Sri Lanka

- Led visual branding for Prakampana '26 flagship event, community empowerment initiatives (Project HOPE), and onboarding logistics (CAST 4).

Skills Summary

Technical Domains: Power Electronics, Power System Optimization, Embedded Systems, IoT, Control Engineering, DSP

Hardware & Embedded: ESP32, Raspberry Pi, NI DAQ, Arduino, Sensor Interfacing, Relay Logic

Software & Simulation: MATLAB, Simulink, LabVIEW, LTSpice, Proteus, SolidWorks, Git

Programming Languages: C, C++, Python, Java, MATLAB Scripting

Languages: English (Professional Proficiency), Sinhala (Native Proficiency)

Soft Skills: Technical Leadership, Team Collaboration, Project Management, Analytical Problem Solving

References

Sen. Prof. Anura Wijayapala

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